

Tarjuman — Speech-to-Text & Text-to-Speech Provider Analysis

Deepgram · OpenAI · Google Gemini | Recommendation for a real-time Arabic→English khutbah transcription app | Prepared June 14, 2026

Bottom line. Your current stack is already the right one — keep it. The best realtime Arabic STT is **Deepgram Nova-3**, and you correctly run a second engine (**OpenAI Whisper**) alongside it for accuracy. The best TTS for your 30+ target languages is **OpenAI**, with the browser **Web Speech API** as the free fallback. Don't add Gemini to the realtime path (it's an LLM doing audio, not a low-latency STT engine). The only changes worth making now are two drop-in model upgrades and a translation retry — and even those come *after* you've field-tested Deepgram in a real masjid. Redundancy is insurance; don't pay the premium before you have users to protect.

Your current architecture (what the advice is built on)

Stage	Provider & model	Role
STT — realtime	Deepgram nova-3 (WebSocket, Arabic, diarize=true , endpointing=500)	Primary live transcript
STT — 2nd opinion	OpenAI whisper-1 (per finalized segment, unhinted)	Accuracy reconcile + off-language detection
Translation	Claude haiku-4.5 / sonnet-4.6	Translate + preserve Islamic terms; reconciles the two STT outputs
TTS	OpenAI tts-1-hd (voice onyx) → Web Speech API	Speaks the <i>translated</i> (usually English) text; browser fallback

You already run **two STT engines** and a **two-tier TTS fallback**. The genuine single point of failure is realtime STT (no Deepgram outage failover); translation also has no fallback.

STT comparison — realtime Arabic transcription

Provider	Realtime streaming	Arabic quality	Diarization (live)	Latency	Price /min	Best at
Deepgram Nova-3	Yes WebSocket	Best. Dedicated monolingual Arabic model, 17 dialects; ~40% lower WER (vendor claim)	Yes (you use it today)	~200–300 ms	~\$0.005–0.007	Accuracy-first realtime Arabic at lowest latency
OpenAI gpt-4o-transcribe / realtime / whisper	Partial realtime via gpt-realtime-whisper ; whisper & gpt-4o-transcribe are batch	Strong (Arabic supported); gpt-4o-transcribe beats whisper	No in realtime (diarize is batch-only)	Tunable, not published	\$0.006 batch · \$0.017 realtime	Best accuracy 2nd-opinion & the natural outage failover
Gemini Live API / 2.5 Flash	Yes, but LLM audio, not STT-grade	OK; transcription is a side-output of an LLM, no word timing guarantees	No	~3–15 s spikes reported	~\$0.037 (token-bundled)	Voice-agent reasoning — not pure transcription

Google's real STT engine is **Cloud Speech-to-Text (Chirp 3)** — a separate product from Gemini, streaming, Arabic-capable, but ~300–600 ms and less Arabic-tuned than Deepgram. Viable, but not better here.

TTS comparison — speaking the translated text (English-primary, 30+ targets)

Provider	Naturalness (English)	Languages	Realtime / latency	Price /10 min*	Best at
OpenAI gpt-4o-mini-tts / tts-1-hd	High; gpt-4o-mini-tts is steerable (tone, pace)	95+ (Urdu, Turkish, Indonesian workable)	Streams; ~1–2 s/segment (fine with prefetch)	~\$0.15 (mini) · ~\$0.27 (hd)	Best all-round: English quality + broad targets + cost
Gemini 2.5 / 3.1 Flash TTS	High, expressive	70+ (best breadth)	~3 s — too slow for live	~\$1.20–1.50	Batch "download the whole session as audio" (future)
Deepgram Aura-2	High, professional	7 only (English + 6 European; no Arabic/Urdu/Turkish/Indonesian)	<90 ms — best latency	~\$0.27	English-only ultra-low-latency niche
Web Speech (browser)	Robotic, OS-dependent	Many (inconsistent)	Instant, on-device	Free	Last-resort fallback (already in place)

*Corrected cost: real speech ≈ 130–150 wpm ≈ ~9K characters per 10 min — not the ~100K some calculators assume. Your "\$~0.15/khutbah" budget is realistic with gpt-4o-mini-tts .

What each provider does best (one line)

Deepgram — STT Fastest, most accurate realtime Arabic with live diarization. The reason your transcript is usable. Keep as primary.	Deepgram — TTS (Aura-2) Lowest latency (<90 ms) but English-only-ish. A niche English fallback, not a fit for your multilingual targets.
OpenAI — STT Best accuracy second-opinion (gpt-4o-transcribe) and the cleanest outage failover (gpt-realtime-whisper) — already wired.	OpenAI — TTS Best all-round : natural English, steerable, 95+ languages, cheapest. Your TTS primary.
Gemini — STT Best for conversational voice agents where an LLM reasons over audio — wrong tool for low-latency transcription.	Gemini — TTS Best language breadth (70+) and expressiveness — but ~3 s latency makes it a batch -only option for you.

The verdict

Speech-to-Text

Primary	KEEP Deepgram Nova-3 (monolingual Arabic) — nothing beats it for realtime Arabic.
2nd opinion	UPGRADE OpenAI Whisper → gpt-4o-transcribe (lower WER, ~same cost, drop-in). Keep the two-engine reconcile.
Outage failover	LATER OpenAI gpt-realtime-whisper — only fires if Deepgram fails to reconnect. Build once you have users.
Skip	SKIP Gemini Live for transcription (latency, no diarization). Google Cloud STT works but isn't better.

Text-to-Speech

Primary	UPGRADE OpenAI <code>tts-1-hd</code> → gpt-4o-mini-tts (steerable, cheaper, 95+ langs). Drop-in.
Fallback	KEEP Web Speech API — free, on-device, already per-item robust.
Niche	OPTIONAL Deepgram Aura-2 only if English dominates and latency complaints appear. Gemini TTS only for a future batch "download audio" feature.

More than one provider — concurrently?

STT: yes, and you already do it right. Deepgram realtime + Whisper second-opinion is a deliberate dual-engine design for accuracy. Keep two — don't add a third concurrent engine.

TTS: no. One cloud primary + Web Speech fallback. Running multiple cloud TTS at once just burns money.

More than one as fallback?

STT: one realtime failover (OpenAI) is enough. **TTS:** the cloud→Web-Speech two-tier is enough.

The gap nobody asked about: translation has *no* fallback — and translation *is* the product. A transient Anthropic blip currently errors the segment. A simple retry / secondary is the cheapest, highest-value resilience win you can make.

What most people miss. Redundancy is insurance, and insurance has a premium — complexity, cost, and forever-maintenance. With zero validated users and a core that hasn't been field-tested in a real masjid, building elaborate multi-provider failover is premature. **Validate Deepgram in the real environment first. Don't build a Deepgram failover before you've proven Deepgram itself works.** An outage today hurts ~0 users; resilience engineering earns its keep once it protects someone.

What I'd actually do — in priority order

1. **NOW** **Field-test Deepgram first.** Everything above assumes the primary is good enough in a real room. Prove that before changing anything (see `docs/field-test-protocol.md`).
2. **CHEAP WIN** **Two drop-in model upgrades:** Whisper → `gpt-4o-transcribe` (STT 2nd opinion) and `tts-1-hd` → `gpt-4o-mini-tts` (TTS). Better quality, lower cost, ~one line each. Do these *after* the field test so you don't change two variables before measuring the first.
3. **CHEAP WIN** **Add a translation retry/secondary.** Translation is the product and has no fallback today. One retry (or a cached/secondary model) is high-value insurance.
4. **WHEN YOU HAVE USERS** **Realtime STT outage failover** to OpenAI `gpt-realtime-whisper`, triggered only when Deepgram can't reconnect.
5. **SKIP FOR NOW** Gemini on the realtime path; Deepgram Aura-2 TTS; any concurrent third engine.

Caveats. Current as of June 14, 2026. Provider capabilities and pricing change frequently; vendor accuracy claims (e.g. Deepgram's "~40% lower WER") are unverified by independent 2026 benchmarks. "Gemini" here distinguishes the Gemini Live API (LLM audio) from Google Cloud Speech-to-Text (Chirp 3, a separate dedicated STT). Costs are estimates for a ~10–20 min khutbah and will vary with speech density and target language.

Key sources:

Deepgram Nova-3 Arabic & pricing & diarization docs (deepgram.com) ·
Deepgram Aura-2 TTS & languages (deepgram.com/product/text-to-speech) ·
OpenAI speech-to-text, realtime transcription & gpt-realtime-whisper
(developers.openai.com) · OpenAI gpt-4o-mini-tts & TTS pricing
(developers.openai.com) · Google Gemini Live API & audio docs & pricing

(ai.google.dev) · Google Cloud Speech-to-Text / Chirp 3
(cloud.google.com/speech-to-text) · Gemini 3.5 Live Translate launch
(marktechpost.com, Jun 2026) · TTS/STT latency & pricing roundups 2026
(smallest.ai, costgoat.com, buildmvpfast.com).